

House Price Prediction Using Machine Learning

By: Shubham Meena

Objective

The objective of this project was to develop a machine learning model capable of predicting house prices based on property characteristics such as area, number of bedrooms, bathrooms, parking spaces, stories, and other amenities. The project aimed to identify the key factors influencing house prices and evaluate the performance of different regression models.

Models Used

The following regression models were implemented and evaluated:

- Linear Regression
- Random Forest Regression

Model performance was assessed using evaluation metrics such as R^2 Score, MAE, and RMSE.

Results

Both models demonstrated reasonable predictive performance on the dataset. Among them, Linear Regression achieved the best results with an R^2 Score of approximately **0.65**, indicating that the model could explain around **65% of the variation in house prices**. The model also produced lower prediction errors compared to Random Forest Regression.

Key Findings

- **Area** was identified as the most influential factor affecting house prices.
- Other important features included **bathrooms, parking spaces, air conditioning, number of stories, and main-road access**.
- House prices showed a **right-skewed distribution**, with most properties concentrated in the mid-price range and a few high-value outliers.
- Properties with **direct main-road access** generally commanded higher prices than similar properties without such access.
- Feature importance analysis and correlation analysis consistently highlighted **area, bathrooms, air conditioning, parking, stories, and main-road connectivity** as major price drivers.
- Linear Regression outperformed Random Forest Regression on this dataset, achieving a higher R^2 score and better overall prediction accuracy.

Conclusion and Recommendation

Based on the analysis real estate businesses should focus not only on the size of a property but also on features that add value, such as main-road access, adequate parking, air conditioning, and additional bathrooms. The analysis showed that houses with better connectivity and amenities often achieved higher prices, even when their area was similar to other properties. Therefore, developers and sellers should highlight these features in marketing campaigns and prioritize them when planning new residential projects to maximize property value and attract more buyers.